

[Home](#) › [My Courses](#) › [Financial Markets](#) › [M3: Correlation](#) › Lesson Notes

Lesson Notes

MODULE 3 | LESSON 3

EXCHANGE-TRADED FUNDS

Reading Time	60 minutes
Prior Knowledge	Bonds, Stocks, Diversification
Keywords	Benchmark, Passive Management, Active Management

In the previous lesson, we examined the role that diversification played in shaping a portfolio's standard deviation. In this lesson, we look at a security that has some diversification built into it: exchange-traded funds. We'll discuss their characteristics, their pros and cons, and the difference between passively managed ones and actively managed ones.

1. Stocks vs. Exchange-Traded Funds

Rather than thinking of which stock to pick, there are benefits to picking multiple stocks. Likewise, instead of picking a single cryptocurrency, it makes sense to pick a number of them. Why? In a single word, **diversification**. We have seen that adding stocks to a portfolio can help significantly lower its variance while keeping the expected return around similar levels. The construction of these portfolios is the topic of an entire course, Portfolio Management, which comes much later in the program. For now, though, is there a more convenient way we can start with a portfolio of securities, rather than a single stock? In this lesson, we'll discuss just such a product: exchange-traded funds.

As the name indicates, they trade on exchanges (as opposed to trading on over-the-counter markets). They are funds in the

sense that they own a collection or portfolio of securities. However, each ETF is considered a single security.

2. ETFs: An Introduction

Exchange-traded funds (ETFs) are baskets or collections of securities. The first ETF was launched nearly 30 years ago, the SPY, and its goal was to track the Standard & Poor's 500 Index, known more commonly as the **S&P500**. These represent the largest 500 stocks by market capitalization. Its ticker is **SPY** and it still exists today.

Initially, ETFs were created to follow benchmarks like the NASDAQ-100 (QQQQ) and the Dow Jones Industrial Average (DIA). The idea was to make broad market coverage available in a convenient, low-cost way. If someone only had \$10,000 to invest in stocks, they likely could not afford to purchase 100 shares of the stocks on any of these indices. It was not possible to buy fractional shares of stocks. So, one would have to choose among many possibilities of stocks. Broadly owning the market avoids the problem of stock selection and instead allows for participation in the general market.

ETFs have a price that is given by their **net asset value**. The net asset value, or NAV, is defined as:

$$\text{NAV} = \frac{\text{Assets} - \text{Liabilities}}{\text{ETF Shares Outstanding}}$$

For a traditional equity ETF, the NAV is usually calculated every day, at the end of day, when all the markets being tracked by a particular ETF's index have closed. If the ETF is tracking U.S. equities, then the NAV is calculated after the U.S. markets close at 4:00pm ET. However, there are ETFs that trade in the U.S. but track a European index, say the FTSE 100 in England, and the London Stock Exchange closes at 11:30am ET. As such, the ETF's NAV is calculated shortly after 11:30am, but the ETF keeps trading in the U.S. until 4:30am. In this instance, we could say that the NAV is *stale*, and it is possible that the price at which an ETF trades in the U.S. is different from the NAV between the hours of 11:30am and 4:00pm ET.

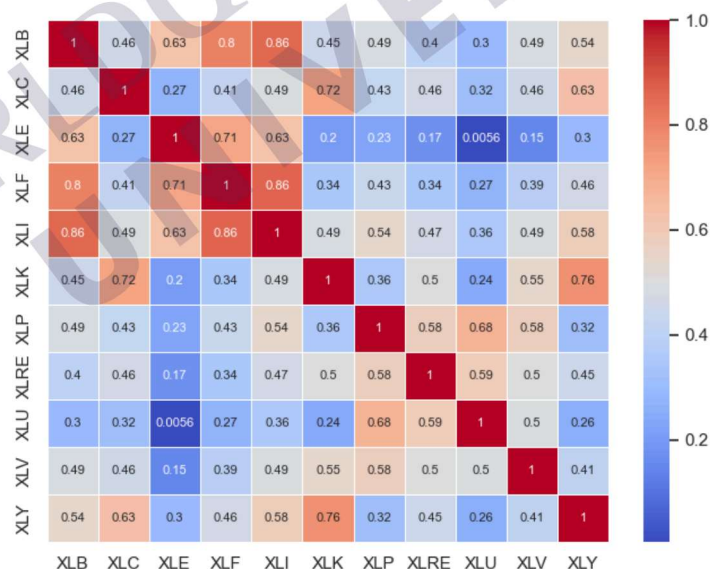
Over time, market makers realized the popularity of these indices and began to create additional ETFs that focused on individual sectors. Indeed, there are 11 sectors that comprise

the S&P500. Each of these sectors has a corresponding ETF. The first round of these were called select SPDRs.

These are:

1. XLC Communication Services
2. XLY Consumer Discretionary
3. XLP Consumer Staples
4. XLE Energy
5. XLF Financials
6. XLV Health Care
7. XLI Industrials
8. XLK Information Technology
9. XLB Materials
10. XLRE Real Estate
11. XLU Utilities

Figure 1: Correlation Matrix of the 11 S&P500 Sectors (2021)



A correlation matrix displays all the correlation coefficients between multiple variables. The diagonal values are always 1.0, representing perfect self-correlation, while off-diagonal values range from -1.0 to 1.0. In Figure 1, notice that the lowest correlation (0.56%) is found between utilities (XLU) and energy (XLE). Note that many utility companies have their prices regulated by state and local governments, so their prices do not necessarily follow market trends the way other sectors do. This

is also shown by having a substantially lower correlation to the overall market than any other sector does (Bausys). All other pairs have some amount of positive correlation, as high as 86% correlation for Industrials and Financials and the same high correlation for Industrials and Materials.

Other market makers and financial institutions have created their own versions of sector-specific ETFs. Some of these tickers are XSW, RYT, VGT, XSD, IGM, FTXL, and QTEC. These are issued by such companies as Invesco, Vanguard, Fidelity, State Street, and iShares, among others.

Consider the select SPDR specializing in technology, XLK. Its top 10 holdings as of 06/10/2024 are:

Symbol	Company Name	Index Weight
MSFT	Microsoft Corp	22.58%
AAPL	Apple Inc.	21.05%
NVDA	Nvidia Corp	5.87%
AVGO	Broadcom Inc.	4.76%
AMD	Advanced Micro Devices	2.64%
CRM	Salesforce, Inc.	2.39%
QCOM	Qualcomm Inc	2.38%
ADBE	Adobe Inc.	2.12%
ORCL	Oracle Corp	2.03%
AMAT	Applied Materials, Inc	1.94%

Within this, 22.58% of the fund is held within MSFT stock, and about 2% is in Applied Materials, Inc. In the next section, we'll discuss benefits of ETFs.

3. ETF Benefits

The first benefit of ETFs is simply the ability to access broad market diversification. In Lesson 2 of this module, we learned that there are benefits to having a diversified portfolio. Essentially, owning a select SPDR gives a broad diversification across a particular industry. Determining which technology company to invest in is certainly more difficult than investing in the technology sector as a whole. The XLK gives broad market exposure with the purchase of a single security. Here is why it

makes sense again to distinguish securities from portfolios: technically, the ETF is a single security, but within it, there is a portfolio of other securities. From a small investor's perspective, however, it provides one of the only mechanisms by which they can own a percentage of every technology company in the S&P500. There are even ETFs that specialize in cryptocurrencies. As such, market exposure could actually mean different securities within a specialized market like cryptocurrencies. Examples of ETFs that specialize in cryptocurrencies are BITW, BLOK, BLCN, LEGR, and KOIN.

A second benefit is professional management. Some ETFs have full-time managers who actively make decisions. Other ETFs have full-time managers who strive to track a benchmark. The decisions they make are more passive, in the sense that they are not trying to beat the market, or benchmark, but rather to track it. Either way, it is prudent to have a trained and dedicated professional managing the complexities of the securities comprising the ETF. If any investor, small or large, were managing the portfolio with stocks, they would likely be spending a lot of time deciding when and how to re-balance the portfolio.

A third benefit is trading flexibility. Some people compare ETFs to open-ended mutual funds. An open-ended mutual fund looks a lot like an ETF, but the open-ended mutual fund has one drawback: you can only buy or sell once a day, at the closing price. If there are market moves that occur during the day that cause you to want to sell, your sell order only gets processed at the closing price. A lot can happen between when you submit your sell order and the closing price. ETFs avoid that problem because they trade like stocks. That is, they trade on exchanges (which is part of their name); they can be bought or sold short throughout the day. This flexibility includes the fact that ETFs can be shorted. If an investor believes a sector to be overvalued, then that investor can sell short the ETF, wait for the price to drop, and cover (buy back) the ETF at the lower price, achieving a capital gain.

The fourth benefit is tied to the third one. ETFs tend to have low fees. The transaction costs, from commission to market impact, tend to be relatively small for ETFs. Surprisingly, not all 500 stocks have similar market impact. The top 50 stocks, say, are much easier to trade in terms of having better execution or

less market impact than, say, the bottom 100 stocks of the S&P500.

4. ETF Disadvantages

Keep in mind that ETFs are not without their disadvantages. There are several main disadvantages.

Like other exchange-traded products, ETFs require investors to trade through a broker that charges commissions and other fees. These fees do not exist for the mutual fund buyer. Buyers of mutual funds typically buy directly from the fund itself. Also, if there is not much trading activity for a particular ETF, it may show up as having a wide gap between the price at which you buy the ETF and the price at which you sell the ETF. This is known as the bid-ask spread, which will be discussed in a later module in this course. Some ETFs are much more popular than others, so this liquidity problem will only exist for some names.

ETFs may have high operating expenses. Each ETF has an expense ratio. This expense ratio is fixed regardless of the fund's performance. If the fund were flat, the expense ratio would mean the net return on the investment is negative. If the ETF were negative on the year, the expense ratio would make it more negative. Prudent investors select a good combination of superior performance and reasonable fees.

ETFs often set out to track a particular benchmark. However, given the inexperience or poor performance of management, the ETF may fail to do so. ETFs can and do deviate from the very benchmark or index they attempt to track. Tracking errors can result in not only higher expense fees but also missed opportunities that create significant underperformance.

ETFs can also suffer from a lack of diversification. The select SPDRs we've discussed are each concentrated within a single industry. So although these ETFs provide diversification against idiosyncratic risk of individual companies, they are still very much exposed to industry risk.

ETFs also have an unusual effect on stocks. The more ETFs that include a stock among its constituency, the greater the demand for that stock (among the ETF portfolio managers). However, the converse is true. If stocks are no longer part of an index or perhaps receive a credit downgrade, then they may

have much less demand. Such actions can cause ETFs to have volatility around index re-balancing.

To understand the pros and cons of ETFs, it helps to divide them into two categories: passively managed ETFs and actively managed ETFs.

5. Passive Management

ETF managers seek to track an index. There is a great amount of transparency in these funds, as their constituency mirrors the weights and members of the index it tracks. Due to the stability of indices, the ETFs do not need to re-balance or turn over often. This stability results in both lower transaction costs and a lower management fee. Similarly, because of low turnover, there are not many capital gains.

Who buys passive funds? In today's market, it helps to think of two types of investors. The first group are traditional investors: small investors wanting broad market exposure who would otherwise not own a wide portfolio of stocks. The second group are robo-advisors: algorithmic trading systems that use artificial intelligence to make investment decisions.

Robo-advisors, one of the most well-known FinTech areas, disrupt the market share and profitability of the portfolio management world. These bots determine when to enter a particular sector, how long to hold it, and when to exit. Providing low-cost and strategic decision-making, robo-advisors are among the largest investors of ETFs. Most (not all) robo-advisors exclusively focus on passive ETFs for their low costs and broad exposure. The value the robo-advisor adds is their expertise in timing the market. Typically, this strategy is known as sector rotation. Robo-advisors use AI to gauge the economy, the markets, and even the individual's specific risk, cash flow, and tax situation to create portfolios that maximize returns and/or income and minimize costs and/or taxes.

6. Active Management

Compare this to actively managed funds. An actively managed ETF seeks to outperform the competition, whether it is a benchmark, an index, or the other ETFs in its category. Active managers hire specialized teams, build infrastructure, and perform analytics, all incurring higher costs. For this extra fee,

they are expected to earn a higher rate of return. While active managers often have a benchmark to beat, they have liberty to deviate as well under different market conditions. Clearly then, the success of actively managed funds depends heavily on the experience and skill of the manager. While their fees are certainly higher than those for passive funds, their returns may not be. Not all active managers outperform the market. Indeed, an active manager who has underperformed the market has both higher costs and lower returns. Surprisingly, only 1% of active ETF managers outperformed the S&P 500 (Chen).

Figure 2: Expense Ratios of ETFs



*The expense ratios of passive ETFs are shown in purple while the ones for active ETFs are shown in green.

7. Sector Selection vs. Stock Selection

What are the differences between picking individual sectors vs. individual stocks? Sector selection tends to involve macro-economic modeling and an understanding of business and economic cycles. It involves the study of monetary policies, employment trends, housing markets, and foreign exchange models and considers the big picture. By avoiding a focus on specific companies, sector selection seeks to determine the best combination of market and sector and looks to rotate among sectors and regions. Once sectors are identified, there are several choices of ETFs from which to choose.

Compare that to stock selection. Given the sheer number of companies, stock selection has more choices than sector selection or a specific ETF sector selection. Once you pick a company, you may compare it to a peer (relative analysis) or to the industry at large. The stock will inevitably have idiosyncratic

risk because anything bad happening to the company will affect that stock price, without necessarily affecting the overall market.

As you will see in a later module, stock selection can be done using fundamental analysis, technical analysis, behavioral analysis, sentiment analysis, or statistical analysis. Selecting stocks seems to be a more complex problem than selecting sectors. Nevertheless, the reward for selecting the stock can be greater, as choosing the right stocks or cryptocurrencies can achieve returns in the thousands of percent that are simply implausible for sector-wide securities.

8. Conclusion

To date, we've discussed four asset classes: bonds, equities, cryptocurrencies, and ETFs. Technically, ETFs can simply contain securities from the first three asset classes, but for our program, we will consider them as a separate asset class. This module built up the importance of diversification, and this lesson showed how ETFs can help achieve that. In the next lesson, we'll examine what happens when the market turns downward. What happens when the market collapses?

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